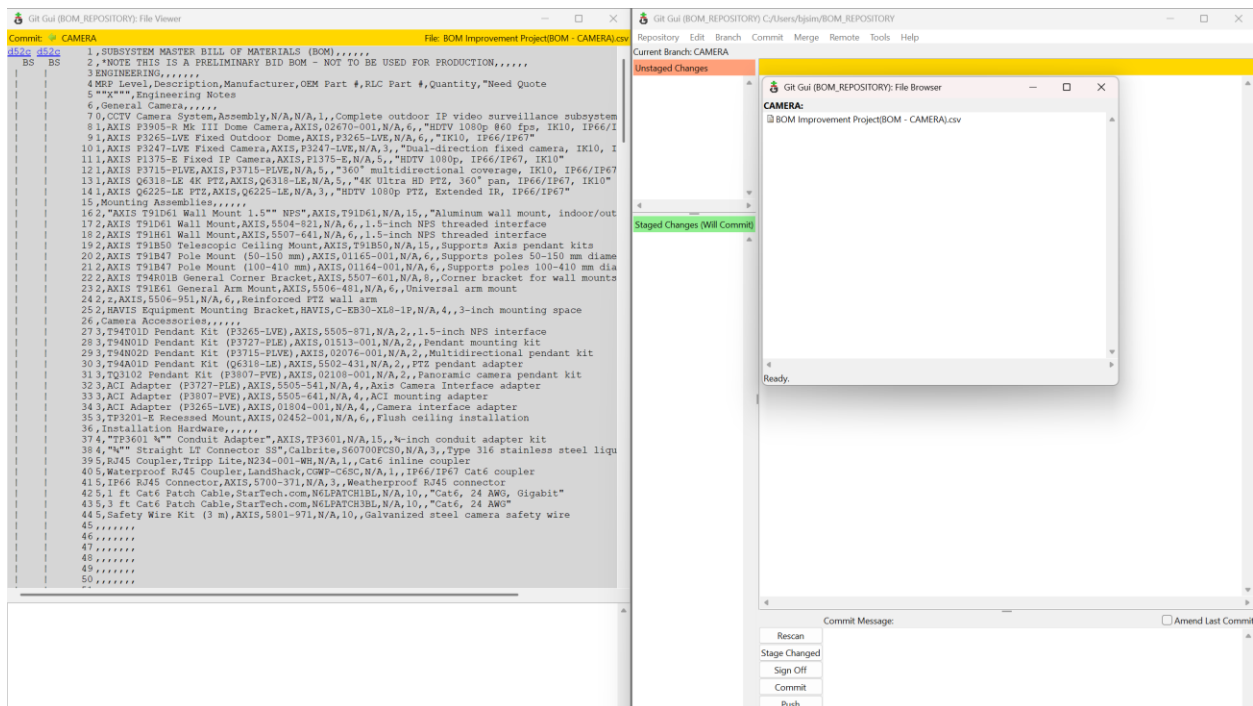


Repository available on: <https://github.com/bsimpsonRL/BOM-Repository>

Introduction:

Utilizing GIT as a configuration management repository, the reference bill of materials comprising security, radio, camera, and network materials can be easily accessed and changed by all individuals seeking to make improvements or get parts from. Through GIT, the reference BOM is divided into four separate branches, with them being the security, radio, camera, and network sections of the overall BOM. Inside each of these branches is a CSV file detailing the specific materials included. These CSV files can be accessed through GIT. For example, accessing the CSV file for the section of the BOM comprising camera materials looks like this:



Here, I am accessing the branch for the camera materials by going into the “Branch” tab on the GUI and clicking “Checkout” of the master branch. Then, I am going into the branch for the camera materials by selecting it. I can then see the repository by clicking the “Repository” tab on the GUI. This can be done for the branches for network, security, and radio. Materials inside each branch can be accessed through this method.

Modifying BOMs:

In the case that new materials or sections need to be added to the repository, the user must use GIT Bash in order to update the CSV files and add new branches to make room for new files. Using the .xlsx file that is included in the master branch of the repository, changes can be made to this spreadsheet that will reflect across the whole repository. However, you must use the “git add” command on Bash to update these changes and then use “git commit -m” followed by a message to fully commit the changes. To add new branches, you can create a branch through GIT Bash by using the “git branch -d” command followed by a name for the branch. This can also be done through the GUI. Files can be added to these branches by entering them through Bash by using the “git add” command and can also be done through the GUI by simply putting them into the master folder for the repository.

Accessibility:

The repository on GitHub can be accessed and changed by all people with capability to enter the repository. To access the repository, a person must first install Git and have an account on GitHub. Then, the command “git clone <https://github.com/bjsimpsonRL/BOM-Repository.git>” can be run on Git Bash to clone the repository to their computer. The repository can also be accessed online through GitHub, but to make changes, the repository must be installed. To make changes to the BOM, the same commands can be applied as in the “Modifying BOMs” section, but to publish these changes to the GitHub repository the command “git push” must be run through Bash. To account for changes that other people may make, it is important to always run “git pull” before making any changes. This command pulls any other changes that people have made to the past to your working version of the repository.

Ideally, changes made to a specific BOM will be made on the branch that the BOM resides on. For example, changes to the BOM for camera materials should be made on the branch within GitHub for this BOM. Instead of having all changes be made on the main branch, this path works better to keep workload flow easy to understand and to avoid cluttering the repository with untraceable changes. For the master Excel file included in the repository, Git can track that the Excel file changed, but it can't reliably merge two people's cell-level changes in an .xlsx file. For this, it is important that category-specific changes are made to the CSV files included in each branch.